



Samsung's 32-litre solo microwave oven with food warming

Microwaves are also required to meet emission standards, which say these cannot emit more than five milliwatts of radiation per square-centimetre beyond 5.08 centimetres (2-inch) from the unit. This prevents any potential damage to pacemakers.

Benefits of using a microwave oven

Microwave ovens only use electricity—not gas, charcoal or oil—which eliminates pollution in the vicinity. Dangerous organisms have less time to thrive on food that is cooked in the microwave because the food is brought to a high temperature quickly and can be served right away.

Microwave ovens can reduce the amount of vitamins B, B1, C and E in foods. It can also damage the protein and amino acid contents in meats and reduce the food's nutritional value. However, it is often better than boiling the same foods, which will remove even more of these nutrients because nutrients are lost through prolonged exposure to heat.

Energy usage rates by a microwave oven typically range between 750 watts and 1100 watts. This is higher than a refrigerator but lower than a clothes iron. The kilowatt-hour per unit usage ranks at about 200kWh, which is more than a colour TV. You can be relaxed while cooking with a microwave oven—just place

the food inside, set the time, stir in between, if necessary, and take the food out once it is done.

Difference between convection oven and OTG

Some of the differences are given below.

Heating method. An OTG can be used for cooking food that is crispy and juicy like *kebabs*, *meats* and *tikkas*. It cooks food with heated coils. When cooking with an OTG, heat is radiated by the coils and absorbed by the food. There is a fan present to distribute the heat, but most of the heat is dispensed directly below the coils. A convection utilises microwave with a fan for heating purpose.

Capacity. An OTG's capacity may go up to 60 litres, whereas a convection microwave's capacity is only up to 32 litres. So, for large batches of food, an OTG might be the better option.

Additionally, rotating surface in a microwave is quite small, which makes putting large dishes in it a problem.

Preheating. Where an OTG can take up to 15 to 20 minutes to preheat, the convection oven only takes around five minutes. This can considerably decrease cooking time.

Cost. While the upfront cost of an OTG is lower, energy efficiency of a microwave oven is higher. A

microwave oven uses 20 per cent less energy than an OTG, saving money in the long run.

A microwave has a digital convection display as compared to an OTG, which has a manual knob.

The dos and don'ts

Never try to repair the microwave on your own. If the grid on the back of the glass has started rusting and peeling away, if the interlocks do not work or the machine gives you any reason to think it might be leaking microwaves, get it repaired or replaced straight away.

A microwave oven should not be operated if the oven door is damaged or there are problems with closing it securely. Do not boil water in a microwave, as it can be superheated in a microwave oven to the point where it will erupt violently and cause burns. Liquids heated inside a microwave oven should be handled cautiously.

Metal objects should not be placed inside a microwave oven, unless these are designated as safe for use. Placing unsafe objects in a microwave may cause sparking inside the oven cavity, which can damage the oven. A microwave oven should be cleaned regularly and not be operated while empty. Use microwave-safe bowl, mugs and containers only, as certain plastic containers can melt and get ruined.

There are also some foods that you should not microwave. These include eggs, chilli peppers, fruits, breast milk, processed foods, leafy greens, oils like olive, peanuts, avocado, etc. Cooking rice in a microwave is also not a good idea because rice has certain bacteria that are killed only when it is boiled on the stove. Breast milk has certain immunological properties that are destroyed in the heating process. However, cow's milk and baby bottles can be microwaved.

Although a convection oven has better features, it will not be of any use if you only want to use it for basic things. So, select one on the basis of what you want to cook. **EFY**